Dynamic Beam Hopping Resource Allocation Algorithm Based on Deep Reinforcement Learning in Multi-Beam Satellite Systems

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***Abstract*—Beam hopping technique provides a foundation for the flexible allocation and efficient utilization of resource in multi-beam satellite systems. The beam hopping time plan is a time-sliced transmission plan for dynamic resource allocation. In order to minimize the transmission delay of packets in multibeam satellite systems, the optimization problem is formulated and a beam hopping resource allocation algorithm based on deep reinforcement learning is proposed in this paper. Firstly, the forward link traffic model of multi-beam satellite systems is established, in which the satellite is modeled as an agent. Then the beam hopping pattern based on interference avoidance criterion is designed as the agent action set. Finally, the deep reinforcement learning algorithm is used to minimize the transmission delay of packet. Simulation results show that compared with the traditional algorithm, the proposed method can effectively decrease the transmission delay of packet and improve the system throughput.**

Keywords—beam hopping, deep reinforcement learning, dynamic resource allocation, satellite communications

I.Introduction

In traditional multi-beam satellite systems, the power and frequency resources allocated to each beam are relatively fixed. However, due to the non-uniform and time-varying traffic requests among beams, the traditional allocation algorithm cannot meet the traffic requests [1].

Beam hopping (BH) technique is based on time-sliced: only part of the beams is activated to work at the same time slot. BH technique is driven by traffic requests and can greatly improve system resource utilization [2-3]. Anzalchi et al. [4] assessed the performance comparison between beam-hopping systems and the non-hopped systems in Ka-band. For flexible resource allocation in the forward link of BH multi-beam satellite systems, two heuristic algorithms were proposed in reference [5], where the capacity resources were allocated based on the per-beam traffic demands. Lei et al. [6] proposed two novel capacity optimization approaches assuming per-beam signal to interference and noise ratio (SINR) constraints in order to use the satellite resources more efficiently. Han et al. [7] considered the delay fairness for the users among each cell in BH systems, and intended to dynamically schedule the slot number of each cell for realizing the long-term delay fairness. In order to solve the interference between beams in the full-bandwidth multi-beam satellite systems, reference [8] proposed a BH pattern optimization method based on beam clustering, which avoided interference by calculating the beam distance threshold.

On the other hand, deep reinforcement learning (DRL) is one of the most concerned directions in the field of artificial intelligence in recent years. It combines the perception of deep learning with the decision-making of reinforcement learning, and directly controls the behavior of agents through the learning of high-dimensional perceptual input, which provides a way to solve the perceptual decision-making problem of complex systems [9]. Several researches had represented that the DRL method can achieve better performance in the satellite dynamic resource allocation systems. Luis et al. [10] presented an approach based on DRL to allocate power in multi-beam satellite systems. The analysis showed the promising results for DRL to be used as a dynamic resource allocation algorithm. X. Hu et al. [11] proposed a framework based on DRL for dynamic resource allocation in multi-beam satellite systems. The simulations showed that the framework can effectively adapt to satellite time-varying scenarios. Furthermore, reference [12] established the beam hopping time plan (BHTP) optimization problem to minimize the transmission delay in multi-beam satellite systems, and proposed a beam hopping algorithm based on DRL, which can reduce the transmission delay and improve the system throughput.

However, the existing BH resource allocation algorithms based on DRL does not consider the problem of co-channel interference between beams. The interference is inevitable when the activate working beams are adjacent. In order to alleviate the problem of co-channel interference between beams, this paper designs a set of BH pattern based on the interference avoidance criteria. And then a beam hopping resource allocation algorithm based on DRL is proposed. The action of the algorithm is selected from the designed BH pattern. The purpose of the optimization problem in this paper is to minimize the transmission delay of packets in multibeam satellite systems, in which the optimization problem is modeled as a Markov decision process. Firstly, the satellite and the packets traffic among beams are modeled as the agent and the environment, respectively. At each time slot, new packets traffic will generate among the beams. Then the satellite agent will determine which beams to work according to the traffic information, the on-board power constrained and the channel condition. Finally, the packets among beams will

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be transmitted according to the satellite agent action. Thus, the transmission delay of the packet can be minimized through the continuous interactive learning between satellite agent and environment. Simulation results show that, compared with the traditional algorithms, the beam hopping resource allocation algorithm based on DRL can decrease the transmission delay of packets and improve the throughput of the satellite systems.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Forward Link Traffic Model

The paper considers the forward link of the broadband multi-beam geostationary satellite systems, where the BH technique is applied. K activated beams are illuminated the $N(K \le N)$ cells, in which the activated beams share the spectrum resource of the satellite broadband W.

The N cells are expressed as $\psi = \{c_n \mid n = 1,2,3,...,N\}$. And the beam hop time period is expressed as $\mathcal{T} = \{t_1, t_2, ..., t_j, ...\}$, where the t_j is a beam hopping slot. The considered forward link traffic model is shown in Fig. 1.

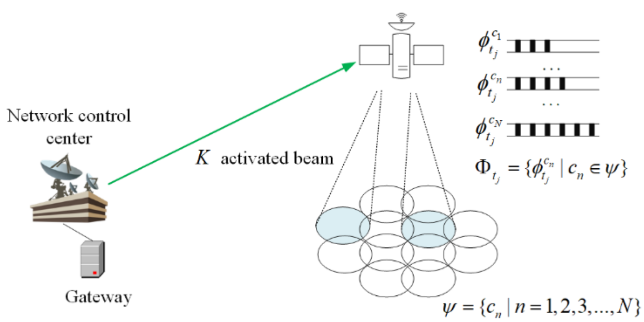

Fig.1 Forward link traffic model

The BHTP is the BH pattern in the beam hop period $\mathcal{T}$. The BH pattern at time slot t_j can be denoted as $X_{t_j} = [x_{t_j}^{c_1}, x_{t_j}^{c_2}, ...x_{t_j}^{c_N}]$, where the $x_{t_j}^{c_n} = 1$ means cell c_n is illuminated at time slot t_j and the $x_{t_j}^{c_n} = 0$ means cell c_n is not illuminated at time slot t_j. According to the BH pattern, the SINR of the illuminated cell c_n at time slot t_j can be calculated as below

$$r_{t_j}^{c_n} = \frac{h_{t_j}^{c_n,c_n} \cdot P_{t_j}^{c_n}}{N_0 W + \sum\limits_{x_{t_j}^{c_i}=1, c_i \in \psi, c_i \ne c_n} h_{t_j}^{c_n,c_i} \cdot P_{t_j}^{c_i}} \tag{1}$$

where $h_{t_j}^{c_n,c_i}$ [12] represents the power gain from activated beam c_i to cell c_n at time slot t_j, which is determined by the satellite transmitting antenna gain, the free space loss, rain attenuation and the receiving antenna gain. $P_{t_j}^{c_n}$ is the satellite transmitting power to cell c_n at time slot t_j. N_0 is the noise power spectral density. The offered capacity for the c_n at time slot t_j can be calculated as below

$$C_{t_j}^{c_n} = W \cdot f_{DVB-S2}(r_{t_j}^{c_n}) \tag{2}$$

Where f_{DVB-S2} in equation (2) is a step function according to the European Telecommunications Standards Institute [13].

The packet arrival rates among cells can be expressed as the Poisson process, which is denoted as $\Lambda_{t_j} = \{\lambda_{t_j}^{c_n} \mid c_n \in \psi\}$. The $\lambda_{t_j}^{c_n}$ means the packet arrival rate for cell c_n during the period $[t_j, t_{j+1}]$. All the packets are stored in the satellite buffer $\Phi_{t_j} = \{\phi_{t_j}^{c_n} \mid c_n \in \psi\}$, where the $\phi_{t_j}^{c_n}$ is the set of packets that need to be transmitted to cell c_n at time slot t_j. Thus, the $\phi_{t_j}^{c_n}$ can be denoted as $\phi_{t_j}^{c_n} = \{pa_{t_j}^{c_n}, pa_{t_{j-1}}^{c_n}, ...pa_{t_{j-Tth}}^{c_n}\}$, where the $pa_{t_k}^{c_n}$ means arrived number of packets for c_n at time slot t_k. The maximum delay time of packet is T_{th}. When the transmission delay of packet $t_{delay}^{j,k} > T_{th}$, the packet will be dropped. If the $pa_{t_k}^{c_n}$ is transmitted at time slot t_j, the transmission delay of packet can be calculated as below

$$t_{delay}^{j,k} = t_j - t_k \tag{3}$$

B. Problem Formulation

During the period $[t_j, t_{j+1}]$, the current satellite buffer Φ_{t_j} depends on the previous state $\Phi_{t_{j-1}}$, packet arrival rates Λ_{t_j}, the channel condition H_{t_j}, the satellite transmitting power P_{t_j} and the BH pattern X_{t_j}. Thus, the satellite buffer at time slot t_j is calculated as given below

$$\Phi_{t_j} = f(\Phi_{t_{j-1}}, \Lambda_{t_j}, H_{t_j}, P_{t_j}, X_{t_j}) \tag{4}$$

where $f(\cdot)$ is the satellite buffer updating formula.

In order to minimize the transmission delay of the packet, the optimization problem can be formulated as (5)

$$opt.P = \min_{X_{t_j}} \sum_{t_j \in \mathcal{T}} \sum_{c_n \in \psi} \sum_{pa_{t_k}^{c_n} \in \phi_{t_j}^{c_n}} (t_j - t_k) \tag{5}$$

$$s.t. \sum_{c_n \in \psi} x_{t_j}^{c_n} \le K, x_{t_j}^{c_n} \in \{0,1\}, \forall c_n \in \psi, t_j \in \mathcal{T} \tag{6}$$

$$\sum_{c_n \in \psi} P_{t_j}^{c_n} \le P_{tot}, \forall t_j \in \mathcal{T} \tag{7}$$

$$P_{t_j}^{c_n} \le P_b, \forall c_n \in \psi, t_j \in \mathcal{T} \tag{8}$$

$$t_{delay}^{j,k} = t_j - t_k \le T_{th} \tag{9}$$

The aim of the optimization problem is to minimize the transmission delay of packets within a beam hop period. The (6) indicates that the maximum illuminated cell in a single time slot should not exceed the number of activated beams K. The (7) represents that the power of all illuminated beams in each time slot should not exceed the satellite total power P_{tot}, and the (8) means that the power of a single beam cannot exceed the single beam power limitation P_b. The (9) means the delay of the packet should not exceed the maximum transmission delay T_{th}.

III. Beam Hopping Resource Allocation Algorithm Based on Deep Reinforcement Learning

In this section, the algorithm architecture is illustrated first. Then the definition of state, action, reward and the Q_m network are described. Finally, the deep reinforcement learning algorithm implementation is summarized.

A. Algorithm Architecture

The satellite and the packets traffic among beams are modeled as the agent and the environment state information, respectively. The algorithm architecture [11] is illustrated in Fig. 2. The Q_m network is the mapping function of action-value to decide the action decisions. In order to improve the performance of Q_m network, the replay memory and target network Q_t are adopted.

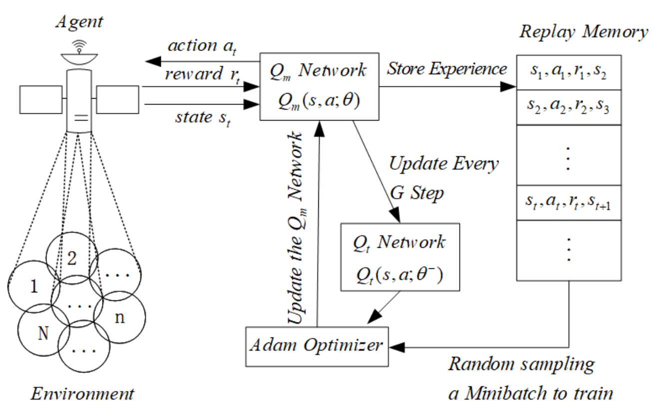

Fig. 2 Deep reinforcement learning architecture

1)State Definition

The state information of the environment depends on the current satellite buffer Φ_{t_j}. Observation includes two kind of information, namely the number of packets W_{t_j} and the average transmission delay D_{t_j}. Thus, the state is defined as

$$s_t = (W_{t_j}, D_{t_j}) \tag{10}$$

Where the W_{t_j} is defined as $W_{t_j} = [w_{t_j}^{c_1}, w_{t_j}^{c_2}, ... w_{t_j}^{c_N}]$. And the $w_{t_j}^{c_n}$ is the number of packets in cell c_n at time slot t_j. The $w_{t_j}^{c_n}$ can be calculated as below

$$w_{t_j}^{c_n} = \sum_{pa_{t_k}^{c_n} \in \phi_{t_j}^{c_n}} pa_{t_k}^{c_n} \tag{11}$$

The average transmission delay is defined as $D_{t_j} = [d_{t_j}^{c_1}, d_{t_j}^{c_2}, ... d_{t_j}^{c_N}]$. And the $d_{t_j}^{c_n}$ is the average transmission delay in cell c_n at time slot t_j. The $d_{t_j}^{c_n}$ can be calculated as below

$$d_{t_j}^{c_n} = \frac{1}{w_{t_j}^{c_n}} \sum_{pa_{t_k}^{c_n} \in \phi_{t_j}^{c_n}} pa_{t_k}^{c_n} * t_{delay}^{j,k} \tag{12}$$

2)Action Definition

When the distance of the activated beams is more than 4 times of the beam radius, the co-channel interference can be ignored [8]. Hence, the BH pattern is designed according to this principle to alleviate co-channel interference. Through iteration, the process of beam hopping pattern sets is shown in Fig. 3. To be noticed that, this paper does not need to get all feasible solutions, one set of feasible solutions is enough to regarded as the action set of the agent. Thus, the action can be defined as below

$$a_t \in Set = \{X_1, ..., X_k, ..., X_{num}\} \tag{13}$$

Where X_k is one of the BH pattern. At each time slot, one of BH pattern in Set is selected and executed.

3)Reward Definition

The goal of the optimization problem is to minimize the transmission delay of packets. On the one hand, according to equation (11) and equation (12), the $w_{t_j}^{c_n} * d_{t_j}^{c_n}$ means the total delay of packet in cell c_n at time slot t_j. Hence, the reward value can be designed to be a big value when the total delay of all packet in cell is small [12], and vice versa. On the other hand, the packet is subject to first-come-first-serve mode. At time slot t_j, the new packet will arrive in cell c_n, and the delay of packet will increase which the packet has not been transmitted before. Thus, the reward can be defined as below

$$r_t = -\| W_{t_j} \odot D_{t_j} \| \tag{14}$$

Where the $\odot$ means the Hadamard matrix operation, and the operator $\|\cdot\|$ means the sum of all elements in a matrix.

4) Q_m Network

The deep neural networks (DNN) is adopted in this paper. Q_m network is the action-value function which evaluates the expected cumulative rewards for taking the action a under the state s [14]. Here, the action-value function is defined as below

$$Q(s,a) = \max_{\pi} E[r_t + \gamma r_{t+1} + ... \mid s_t = s, a_t = a, \pi] \tag{15}$$

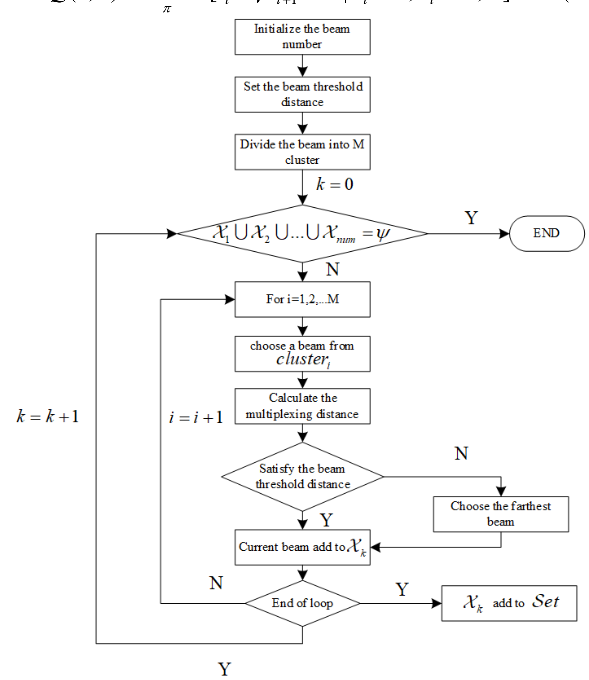

Fig. 3 Process of beam hopping pattern sets

B. Algorithm Implementation

At each time slot, the experience tuple (s_t, a_t, r_t, s_{t+1}) is stored in the replay memory. When the number of experience tuples exceed the replay start step N_{rs}, the Q_m network starts to train. During the training step, a minibatch size N_{mb} are randomly sampled from replay memory. The target network value based on the Bellman Equation [15] is calculated as below

$$\mathrm{y}_t = \begin{cases} r_t & \text{if terminated} \\ r_t + \gamma \max_{a \in A} Q_t(s_{t+1}, a; \theta^-) & \text{otherwise} \end{cases} \quad (16)$$

The loss $L_t(\theta_t)$ of current Q_m network at t_{th} iteration can be calculated as below

$$L_t(\theta_t) = E_{s,a}[(y_t - Q_m(s, a; \theta_t))^2] \quad (17)$$

The main process of the DRL algorithm is summarized in Algorithm 1. The algorithm consists of initialization part and training part. In the initialization part, the satellite system scenario including satellite buffer, Q_m network, Q_t network and replay memory are initialized. In the training part, the Q_m network will make the action decisions and be trained.

Algorithm 1 The DRL algorithm implementation

Initialization
Initialize the satellite scenario and initialize the satellite buffer
Initialize the agent, initialize the Q_m network with random weight θ, Q_t network weight with $\theta^- = \theta$
Initialize the replay memory capacity with N_{rm}
Training
For episode=1, 2, ..., Episode:
  Initialize the satellite buffer $\Phi = 0$
  For $t \in \mathcal{T}$
    Packets among cells arrival
    Observe the satellite buffer and abstract the state into two-dimensional tensor s_t
    If rand $\leq$ ε choose a random action a_t
    Else choose an action a_t with maximum Q value
    End If
    Illuminate the selected beams based on the a_t
    Environment step to next state s_{t+1}, and get the reward r_t
    Store experience tuple (s_t, a_t, r_t, s_{t+1}) in the replay memory
    Random sample a minibatch of (s_t, a_t, r_t, s_{t+1}) from replay memory
    Calculate the label Q value as equation (16)
    Calculate the DNN loss $L_t(\theta_t)$ as equation (17)
    Train the Q_m network in Adam Optimizer
    Every G step, update the Q_t weight $\theta^- = \theta$
  End For
  Decrease the exploration probability ε
End For

IV. Simulation Analysis

A. System Parameters

1) Satellite Scenario Parameters

Herein, a simplified model is considered in which a multi-beam geostationary satellite with user link operating at 20GHz. The system parameters are summarized in Table I. The average traffic demand in each cell is illustrated in Fig. 4.

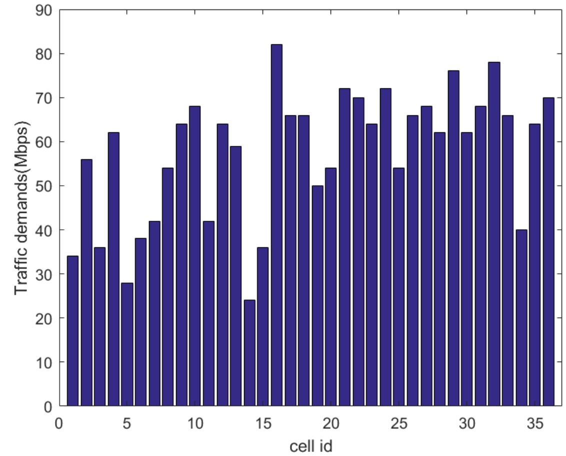

Fig. 4 Traffic demand among each cell

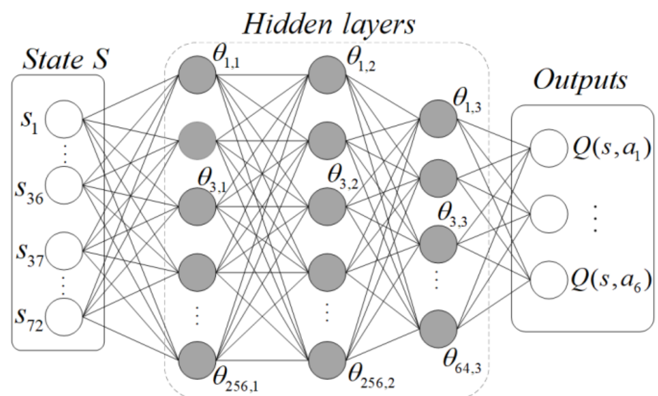

Fig. 5 Q network architecture

2) Alogorithm Parameters

The algorithm parameters are given in Table II. The structure of Q_m network is shown in Fig. 5, where the Q_m network consists of three fully connected layers. The convergence performance of the DRL is validated in Fig. 6.

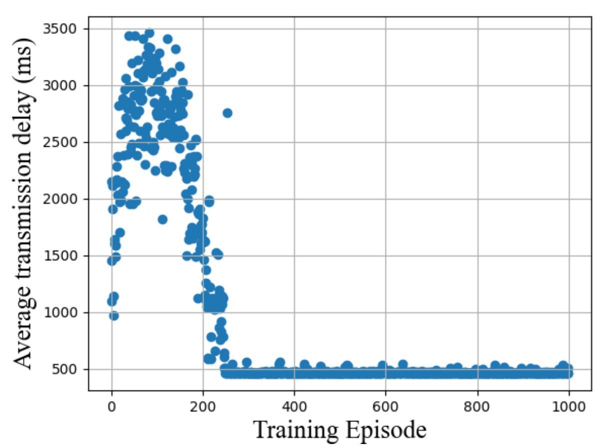

Fig. 6 Average transmission delay during training episode

TABLE I SCENARIO PARAMETERS

Parameters	Illustration	Values
N	Number of cells	36
K	Number of active beams	6
BHP_{num}	Number of BH pattern	6
W	System bandwidth	200MHz
P_{tot}	Total on-board power	300W
P_b	Max single beam power	200W
L_{free}	Free space loss	212dB
L_{rain}	Rain attenuation	4~10dB
G_T	Transmit antenna gain	40dB
G_R	Receive antenna gain	50dB
T_n	Noise temperature	300K
k	Boltzmann's constant	-228.6dB
t_s	Time slot duration	100ms
T_{th}	Packet delay threshold	4s

TABLE II ALGORITHM PARAMETERS

Parameters	Values
Training Episode	1000
Time slot every episode	512
Replay memory size N_{rm}	5000
Minibatch size N_{mb}	16
Replay start step N_{rs}	2000
Main network train frequency	4
Target network update frequency	100
Activation function	Relu
Discount factor	0.90
Learning rate	0.001
Initial exploration rate	0.5
Final exploration rate	0.01

B. Simulation Result and Analysis

In this section, the algorithm performance is evaluated in terms of the average transmission delay of packet and the average system throughput under different normalized traffic. The normalized traffic is defined as the total packets traffic loads among cells divided by the maximum available capacity of the satellite. The RAND algorithm means that the K active beams are randomly selected. And the FixTime algorithm means that fixed time slots are allocated to each beam.

1)Average Transmission Delay Performance

The performance of average transmission delay under different traffic demand is shown in Fig. 7. It is noticed that our proposed solution is significantly better than RAND algorithm and FixTime algorithm, where the results are 453ms, 901ms, 1258ms, respectively. That is to say, compared with the RAND algorithm and FixTime algorithm, the algorithm based on DRL can decrease the average transmission delay in 49.7% and 64.0%.

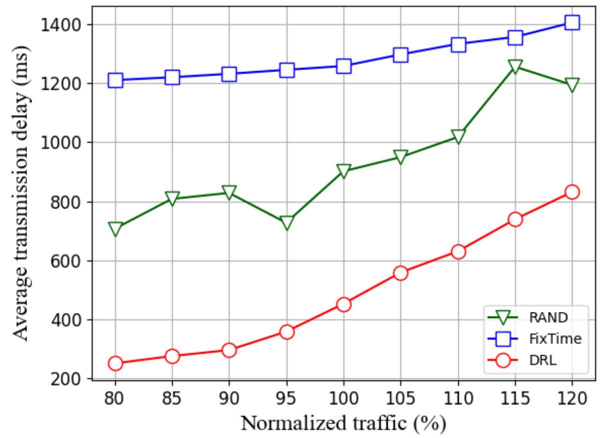

Fig. 7 Average transmission delay under different traffic demand

2)Average System Throughput Performance

The average system throughput is defined as the total packet transmission capacity divided by the number of time slot during a beam hop period. According to the equation (14) of reward and the packet transmission first-come-first-serve mode, when the transmission delay of packet is small, it can be known that the dwell time of packet in the satellite buffer is small. Thus, the shorter the transmission delay of packet, the higher the transmission ability of the satellite. As shown in the Fig. 8, the DRL algorithm can effectively improve the satellite throughput compared with RAND algorithm and FixTime algorithm. When the normalized traffic is 1.0, the average transmission capacity of DRL algorithm, RAND algorithm and FixTime algorithm is 1865Mbps, 1560Mbps and 1214Mbps, respectively. Therefore, compared with the RAND algorithm and FixTime algorithm, the algorithm based on DRL can effectively improve the throughput performance by 19.5% and 53.6%.

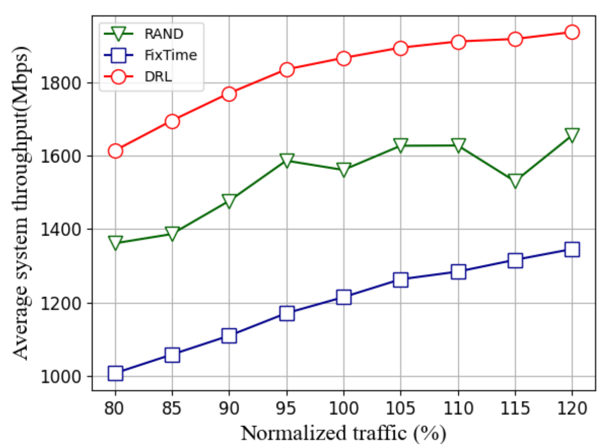

Fig. 8 Average system throughput under different traffic demand

V.CONCLUSION

This paper establishes the traffic model of the forward link in multi-beam satellite systems. And the beam hopping resource allocation algorithm based on DRL is proposed. Then the BH pattern sets is designed as the agent actions to alleviate co-channel interference. The Simulation results show that the proposed algorithm can decrease the transmission delay and improve the system throughput.

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